

Introduction to Python Programming

What is Python?

Python is a high-level, interpreted, and general-purpose programming language created by Guido van Rossum and first released in 1991. Python is known for its simple syntax, readability, and extensive library support.

Features of Python

Easy to learn, interpreted, platform independent, object-oriented, large standard library, open source, dynamic typing.

Variables Example

```
name='John', age=25, salary=35000.50
```

Data Types

int, float, str, bool, list, tuple, dict, set

Conditional Statements

```
if age >= 18: Eligible to vote
```

Loops

```
for i in range(5): print(i)
```

```
while count <= 5: print(count)
```

Functions

```
def greet(name): return f'Hello, {name}'
```

Object-Oriented Programming

Four principles: Encapsulation, Inheritance, Polymorphism, Abstraction

Exception Handling

```
try/except ValueError
```

Python Applications

Web Development, Data Science, AI, Machine Learning, Automation, Cybersecurity, Cloud Computing

Summary

Python was created by Guido van Rossum and first released in 1991.